

**FMUSER FUQ-4459B DVB-C Modulator 8 Tuner Inputs 128 IP
Inputs UDP-RTP 4 RF Outputs
DVB-C-DVB-C-T-S-S2-S2X-ISDB-T-ATSC Multi-Mode Scrambling
BISS Descrambling Web-Based NMS for Broadcast and CATV** Its
production is stopped temporarily, and the date for resuming production is not fixed yet.



Product overview

The FMUSER FUQ-4459D is a DVB-C modulator with 8 tuner inputs, 128 IP inputs via UDP/RTP, and 4 non-adjacent carrier outputs. It integrates multiplexing, scrambling, and modulation functions, supporting BISS descrambling. Managed through web-based NMS, this device is ideal for digital broadcasting networks, enabling flexible signal processing and secure content delivery.

Key feature

Supports multi-mode tuner inputs for DVB-C, DVB-T, DVB-S/S2/S2X, ATSC, and ISDB-T with CAM slots.

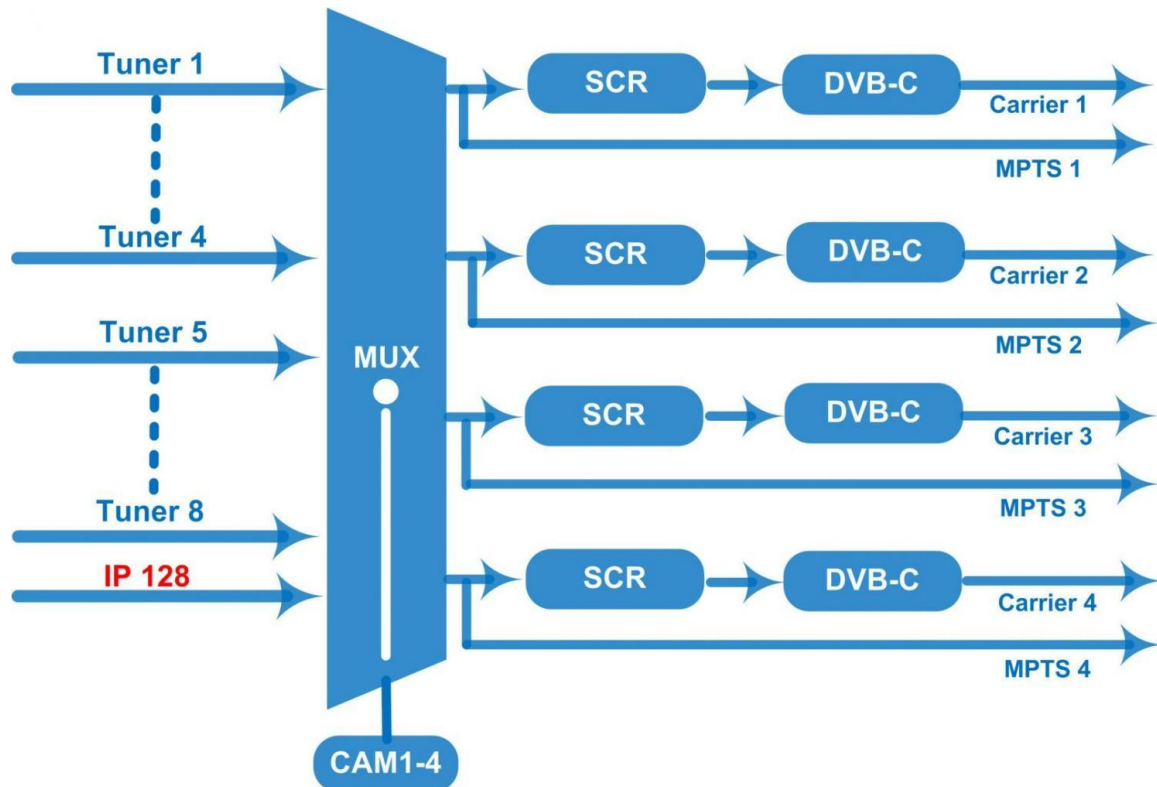
Handles up to 128 IP streams via UDP/RTP protocols.

Features integrated multiplexing, scrambling, and DVB-C modulation for 4 carrier outputs.

Enables BISS descrambling for secure decryption.

Facilitates remote management via web-based NMS.

Principle Chart



Specifications

Input	4 Tuner (DVB-C/T/(T)/S/S2/S2X/ISDBT optional) + 4 Tuner with CAM (DVB-C/T/(T)/S/S2/S2X/AT/SC/ISDBT optional) 128 IP(SPTS/MPTS) input over UDP and RTP, RJ45, 1000M/100M, Unicast/Multicast	4 Tuner (DVB-C/T/(T)/S/S2/S2X/ISDBT optional) + 4 Tuner with CAM (DVB-C/T/(T)/S/S2/S2X/AT/SC/ISDBT optional) 128 IP(SPTS/MPTS) input over UDP and RTP, RJ45, 1000M/100M, Unicast/Multicast	4 Tuner (DVB-C/T/(T)/S/S2/S2X/ISDBT optional) + 4 Tuner with CAM (DVB-C/T/(T)/S/S2/S2X/AT/SC/ISDBT optional) 128 IP(SPTS/MPTS) input over UDP and RTP, RJ45, 1000M/100M, Unicast/Multicast	4 Tuner (DVB-C/T/(T)/S/S2/S2X/ISDBT optional) + 4 Tuner with CAM (DVB-C/T/(T)/S/S2/S2X/AT/SC/ISDBT optional) 128 IP(SPTS/MPTS) input over UDP and RTP, RJ45, 1000M/100M, Unicast/Multicast	4 Tuner (DVB-C/T/(T)/S/S2/S2X/ISDBT optional) + 4 Tuner with CAM (DVB-C/T/(T)/S/S2/S2X/AT/SC/ISDBT optional) 128 IP(SPTS/MPTS) input over UDP and RTP, RJ45, 1000M/100M, Unicast/Multicast	4 Tuner (DVB-C/T/(T)/S/S2/S2X/ISDBT optional) + 4 Tuner with CAM (DVB-C/T/(T)/S/S2/S2X/AT/SC/ISDBT optional) 128 IP(SPTS/MPTS) input over UDP and RTP, RJ45, 1000M/100M, Unicast/Multicast	4 Tuner (DVB-C/T/(T)/S/S2/S2X/ISDBT optional) + 4 Tuner with CAM (DVB-C/T/(T)/S/S2/S2X/AT/SC/ISDBT optional) 128 IP(SPTS/MPTS) input over UDP and RTP, RJ45, 1000M/100M, Unicast/Multicast	4 Tuner (DVB-C/T/(T)/S/S2/S2X/ISDBT optional) + 4 Tuner with CAM (DVB-C/T/(T)/S/S2/S2X/AT/SC/ISDBT optional) 128 IP(SPTS/MPTS) input over UDP and RTP, RJ45, 1000M/100M, Unicast/Multicast
Tuner Section	Multi-mode tuners switchable (New)	DVB-C	DVB-C	DVB-C	Input Frequency	60~890 MHz	60~890 MHz	60~890 MHz
Tuner Section	Multi-mode tuners switchable (New)	DVB-C	DVB-C	DVB-C	Standard	J.83A, J.83C	J.83A, J.83C	J.83B
Tuner Section	Multi-mode tuners switchable (New)	DVB-C	DVB-C	DVB-C	Constellation	16/32/64/128/256 QAM	16/32/64/128/256 QAM	64/256 QAM

Tuner Section	Multi-mode tuners switchable (New)	DVB-T/(T)	DVB-T/(T)	DVB-T/(T)	Input Frequency	60~890 MHz	60~890 MHz	60~890 MHz
Tuner Section	Multi-mode tuners switchable (New)	DVB-T/(T)	DVB-T/(T)	DVB-T/(T)	Bandwidth	6/7/8 M	6/7/8 M	6/7/8 M
Tuner Section	Multi-mode tuners switchable (New)	ISDB-T	ISDB-T	ISDB-T	Input Frequency	60~890 MHz	60~890 MHz	60~890 MHz
Tuner Section	Version 1	DVB-S	DVB-S	DVB-S	Input Frequency	950-2150MHz	950-2150MHz	950-2150MHz
Tuner Section	Version 1	DVB-S	DVB-S	DVB-S	Symbol rate	0.5~45 Msps	0.5~45 Msps	0.5~45 Msps
Tuner Section	Version 1	DVB-S	DVB-S	DVB-S	Signal Strength	-65~-25 dBm	-65~-25 dBm	-65~-25 dBm
Tuner Section	Version 1	DVB-S	DVB-S	DVB-S	FEC	1/2, 2/3, 3/4, 5/6, 7/8 QPSK	1/2, 2/3, 3/4, 5/6, 7/8 QPSK	1/2, 2/3, 3/4, 5/6, 7/8 QPSK
Tuner Section	Version 1	DVB-S2	DVB-S2	DVB-S2	Input Frequency	950 ~2150MHz	950 ~2150MHz	950 ~2150MHz
Tuner Section	Version 1	DVB-S2	DVB-S2	DVB-S2	Symbol rate	QPSK: 1~45 Msps; 8PSK: 0.5~45 Msps	QPSK: 1~45 Msps; 8PSK: 0.5~45 Msps	QPSK: 1~45 Msps; 8PSK: 0.5~45 Msps
Tuner Section	Version 1	DVB-S2	DVB-S2	DVB-S2	FEC	1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10	1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10	1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10
Tuner Section	Version 1	DVB-S2	DVB-S2	DVB-S2	Constellation	QPSK, 8PSK	QPSK, 8PSK	QPSK, 8PSK
Tuner Section	Version 2 (New)	DVB-S	DVB-S	DVB-S	Input Frequency	950 ~2150MHz	950 ~2150MHz	950 ~2150MHz

Tuner Section	Version 2 (New)	DVB-S	DVB-S	DVB-S	Symbol rate	0.5~45 Msps	0.5~45 Msps	0.5~45 Msps
Tuner Section	Version 2 (New)	DVB-S	DVB-S	DVB-S	Signal Strength	- 65- -25dBm	- 65- -25dBm	- 65- -25dBm
Tuner Section	Version 2 (New)	DVB-S	DVB-S	DVB-S	FEC	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Tuner Section	Version 2 (New)	DVB-S	DVB-S	DVB-S	Constellation	QPSK	QPSK	QPSK
Tuner Section	Version 2 (New)	DVB-S	DVB-S	DVB-S	Max input bitrate	≤129Mbps	≤129Mbps	≤129Mbps
Tuner Section	Version 2 (New)	DVB-S2	DVB-S2	DVB-S2	Input Frequency	950 ~2150MHz	950 ~2150MHz	950 ~2150MHz
Tuner Section	Version 2 (New)	DVB-S2	DVB-S2	DVB-S2	Symbol rate	QPSK/8PSK /16APSK :0.5~45 Msps 32APSK: 0.5~40M sps;	QPSK/8PSK /16APSK :0.5~45 Msps 32APSK: 0.5~40M sps;	QPSK/8PSK /16APSK :0.5~45 Msps 32APSK: 0.5~40M sps;

Tuner Section	Version 2 (New)	DVB-S2	DVB-S2	DVB-S2	FEC	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/1 0	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/1 0	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/1 0
						8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/1 0	8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/1 0	8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/1 0
						16 APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/1 0	16 APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/1 0	16 APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/1 0
						32A PSK: 3/4, 4/5, 5/6, 8/9, 9/10	32A PSK: 3/4, 4/5, 5/6, 8/9, 9/10	32A PSK: 3/4, 4/5, 5/6, 8/9, 9/10
Tuner Section	Version 2 (New)	DVB-S2	DVB-S2	DVB-S2	Constellation	QPSK, 16PSK, 32APSK	QPSK, 16PSK, 32APSK	QPSK, 16PSK, 32APSK
Tuner Section	Version 2 (New)	DVB-S2	DVB-S2	DVB-S2	Max input bitrate	≤129 Mbps	≤129 Mbps	≤129 Mbps
Tuner Section	Version 2 (New)	DVB-S2 X	DVB-S2 X	DVB-S2 X	Input Frequency	950 ~2150MHz	950 ~2150MHz	950 ~2150MHz

Tuner Section	Version 2 (New)	DVB-S2 X	DVB-S2 X	DVB-S2 X	Symbol rate	QPSK/8 PSK /16 APSK :0.5~45 Msps 8 APSK/3 2APSK: 0.5~40 Msps	QPSK/8 PSK /16 APSK :0.5~45 Msps 8 APSK/3 2APSK: 0.5~40 Msps	QPSK/8 PSK /16 APSK :0.5~45 Msps 8 APSK/3 2APSK: 0.5~40 Msps
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Tuner Section	Version 2 (New)	DVB-S2 X	DVB-S2 X	DVB-S2 X	FEC	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 8APSK: 5/9-L, 26/45-L 16APSK : 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36 , 2/3-L, 25/36, 13/18, 7/9, 77/90 3 2APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 8APSK: 5/9-L, 26/45-L 16APSK : 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36 , 2/3-L, 25/36, 13/18, 7/9, 77/90 3 2APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9	QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20 8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10 8APSK: 5/9-L, 26/45-L 16APSK : 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36 , 2/3-L, 25/36, 13/18, 7/9, 77/90 3 2APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9
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Tuner Section	Version 2 (New)	DVB-S2 X	DVB-S2 X	DVB-S2 X	Constellation	QPSK, 8PSK, 8APSK, 16APSK, 32APSK	QPSK, 8PSK, 8APSK, 16APSK, 32APSK	QPSK, 8PSK, 8APSK, 16APSK, 32APSK
Tuner Section	Version 2 (New)	DVB-S2 X	DVB-S2 X	DVB-S2 X	Max input bitrate	≤129 Mbps	≤129 Mbps	≤129 Mbps
Tuner Section		ATSC-T	ATSC-T	ATSC-T	Input Frequency	54~858 MHz	54~858 MHz	54~858 MHz
Tuner Section		ATSC-T	ATSC-T	ATSC-T	Bandwidth	6M	6M	6M
Tuner Section		ATSC-T	ATSC-T	ATSC-T	Constellation	8VSB	8VSB	8VSB
Descrambling	CAM/CI	CAM/CI	CAM/CI	CAM/CI	4 sets	4 sets	4 sets	4 sets
BISS Descrambling	Mode 1, Mode E (descramble individual program)	Mode 1, Mode E (descramble individual program)	Mode 1, Mode E (descramble individual program)	Mode 1, Mode E (descramble individual program)	Mode 1, Mode E (descramble individual program)	Mode 1, Mode E (descramble individual program)	Mode 1, Mode E (descramble individual program)	Mode 1, Mode E (descramble individual program)
Multiple xing	Maximum PID Remapping	Maximum PID Remapping	256 per input channel	256 per input channel	256 per input channel	256 per input channel	256 per input channel	256 per input channel
Multiple xing	Function	Function	A/V PID filtering	A/V PID filtering	A/V PID filtering	A/V PID filtering	A/V PID filtering	A/V PID filtering
Multiple xing	Function	Function	PID Remapping	PID Remapping	PID Remapping	PID Remapping	PID Remapping	PID Remapping
Multiple xing	Function	Function	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting
Multiple xing	Function	Function	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically	Generate PSI/SI table automatically

Scrambling	Maximum simulcrypt CA	Maximum simulcrypt CA	4	4	4	4	4	4
Scrambling	Standard	Standard	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197
Scrambling	Connection	Connection	Local/remote connection	Local/remote connection	Local/remote connection	Local/remote connection	Local/remote connection	Local/remote connection
Modulation	DVB-C	DVB-C	Standard	Standard	J.83A (DVB-C), J.83B,	J.83A (DVB-C), J.83B,	J.83A (DVB-C), J.83B,	J.83A (DVB-C), J.83B,
Modulation	DVB-C	DVB-C	MER	MER	≥43dB	≥43dB	≥43dB	≥43dB
Modulation	DVB-C	DVB-C	RF frequency	RF frequency	50~960 MHz, 1KHz step	50~960 MHz, 1KHz step	50~960 MHz, 1KHz step	50~960 MHz, 1KHz step
Modulation	DVB-C	DVB-C	RF output level	RF output level	-25~ -1dbm(83~107 dbμV), 0.1db step	-25~ -1dbm(83~107 dbμV), 0.1db step	-25~ -1dbm(83~107 dbμV), 0.1db step	-25~ -1dbm(83~107 dbμV), 0.1db step
Modulation	DVB-C	DVB-C	Symbol rate	Symbol rate	5.000~7.000Mps adjustable	5.000~7.000Mps adjustable	5.000~7.000Mps adjustable	5.000~7.000Mps adjustable

Modulation	DVB-C	DVB-C	RF Out	RF Out	4*DVB-C non-adjacent carriers combined output, 40M total bandwidth limitation	4*DVB-C non-adjacent carriers combined output, 40M total bandwidth limitation	4*DVB-C non-adjacent carriers combined output, 40M total bandwidth limitation	4*DVB-C non-adjacent carriers combined output, 40M total bandwidth limitation
Modulation	DVB-C	DVB-C	Constellation	Constellation	J.83A	J.83A	J.83B	J.83B
Modulation	DVB-C	DVB-C	Constellation	Constellation	16/32/64/128/256QAM	16/32/64/128/256QAM	64/256QAM	64/256QAM
Modulation	DVB-C	DVB-C	Bandwidth	Bandwidth	8M	8M	6M	6M
Stream out	4*MPTS over UDP and RTP out (RJ45, 1000M/100M Unicast/Multicast)	4*MPTS over UDP and RTP out (RJ45, 1000M/100M Unicast/Multicast)	4*MPTS over UDP and RTP out (RJ45, 1000M/100M Unicast/Multicast)	4*MPTS over UDP and RTP out (RJ45, 1000M/100M Unicast/Multicast)	4*MPTS over UDP and RTP out (RJ45, 1000M/100M Unicast/Multicast)	4*MPTS over UDP and RTP out (RJ45, 1000M/100M Unicast/Multicast)	4*MPTS over UDP and RTP out (RJ45, 1000M/100M Unicast/Multicast)	4*MPTS over UDP and RTP out (RJ45, 1000M/100M Unicast/Multicast)
System	web-based NMS support	web-based NMS support	web-based NMS support	web-based NMS support	web-based NMS support	web-based NMS support	web-based NMS support	web-based NMS support
System	Ethernet software upgrading	Ethernet software upgrading	Ethernet software upgrading	Ethernet software upgrading	Ethernet software upgrading	Ethernet software upgrading	Ethernet software upgrading	Ethernet software upgrading
General	Dimension	Dimension	Dimension	482mm×300mm×44.5mm (W*D*H)	482mm×300mm×44.5mm (W*D*H)	482mm×300mm×44.5mm (W*D*H)	482mm×300mm×44.5mm (W*D*H)	482mm×300mm×44.5mm (W*D*H)

General	Temper ature	Temper ature	Temper ature	0~45■(Operati on) ; -20 ~80■(St orage)	0~45■(Operati on) ; -20 ~80■(St orage)	0~45■(Operati on) ; -20 ~80■(St orage)	0~45■(Operati on) ; -20 ~80■(St orage)	0~45■(Operati on) ; -20 ~80■(St orage)
General	Power	Power	Power	AC 100 V±1050/ 60Hz; AC 220 V±10%, 50/60H Z	AC 100 V±1050/ 60Hz; AC 220 V±10%, 50/60H Z	AC 100 V±1050/ 60Hz; AC 220 V±10%, 50/60H Z	AC 100 V±1050/ 60Hz; AC 220 V±10%, 50/60H Z	AC 100 V±1050/ 60Hz; AC 220 V±10%, 50/60H Z

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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